Precambrian Research 286 (2016) 379–397

Contents lists available at ScienceDirect

Precambrian Research

journal homepage: www.elsevier.com/locate/precamres

A late Paleoproterozoic key pole for the Fennoscandian Shield: A paleomagnetic study of the Keuruu diabase dykes, Central Finland

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ARTICLE INFO

Article history:
Received 12 November 2015
Revised 11 October 2016
Accepted 27 October 2016
Available online 28 October 2016

Keywords:
Fennoscandian Shield
Late Paleoproterozoic
Paleomagnetism
Rock magnetism
Geochemistry

ABSTRACT

We present an updated paleomagnetic pole and a new isotope age for the late Paleoproterozoic (Svecofennian) diabase dykes from Keuruu, Central Finland. The paleomagnetic results reveal a dual-polarity remanent magnetization with asymmetry, which we explain by an unremoved secondary component contaminating both normal (N) and reversed (R) vectors. The ChRM is carried by magnetite, most of which are primary grains crystallized from basaltic magma. The primary nature of the ChRM is further supported by positive baked contact tests. An R polarity dyke shows a U-Pb concordia age of 1870 ± 9 Ma. Zircons from another dyke of N polarity reveal a mean Pb-Pb age of 1867 ± 8 Ma. The combined N and R mean ChRM direction is D = 340.4°, I = 35.2°, α_{95} = 5.8°, k = 31.3, N = 21 dykes (yielding a paleomagnetic pole of Plat = 45.7°, Plon = 230.9°, A_{95} = 5.5°. This positions the Fennoscandian Shield at low latitudes (19.4°), and aligns with Superior Craton in a way that would naturally proceed into the 1.8–1.2 Ga NENA (North Europe–North America) configuration. Paleosecular variation analysis of N polarity dykes is consistent with the low latitude position of Baltica.

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1. Introduction

Paleomagnetism, coupled with precise isotope age data, remains the only quantitative method to reconstruct Precambrian continents to a paleogeographic reference frame. Paleomagnetism provides paleomagnetic poles, which are used to construct apparent polar wander paths (APWP) and to assemble continents into their ancient positions. For this it is necessary to rely on key poles (Buchan, 2013), i.e. poles coupled with a precise age and supported by robust evidence of their primary nature. Only few poles of Baltica between 1.9 Ga and 1.7 Ga are precisely dated and can be regarded as key poles (Pisarevsky and Sokolov, 2001; Elming et al., 2009; Pisarevsky and Bylund, 2010).

The exact position of the Fennoscandian Shield during the Svecofennian orogeny is uncertain, due to a possible discrepancy between the age of crystallization and age of magnetization (delayed by ca. 10–20 Ma) of the slowly cooled Svecofennian (1.88–1.87 Ga) gabbros (Mertanen and Pesonen, 2005 and references therein) upon which position of the Fennoscandian Shield relies. To overcome the problems associated with the slow cooling syn-orogenic intrusions, we turn our focus on the post-orogenic mafic dykes in Finland (Pesonen, 1987). Mafic dykes are ideal for paleomagnetic studies for several reasons. First, they cool rapidly and therefore provide an accurate record of the Earth's magnetic field (e.g. Halls et al. (2007)). Second, they sometimes contain zircon and/or baddeleyite grains which are used to establish the precise crystallization ages of the dykes (e.g. Halls et al. (2007), French and Heaman (2010)). Third, the primary nature of the ChRM in dykes can be established by means of a baked contact test (Everitt and Clegg, 1962; Buchan, 2013).

Earlier paleomagnetic (Pesonen and Neuvonen, 1981; Palmu, 1982; Pesonen et al., 2012) studies have been carried out on early Proterozoic diabase dykes of Keuruu, central Finland. Paleomagnetic results from some of the dyke outcrops show a dual polarity characteristic remanent magnetization (ChRM) with asymmetry between normal and reversed directions. Pb-Pb isochron ages were measured for different combinations using Pb isotope ratios for total rock, titanite, zircon, 3.2–3.4 g/cm^3 mineral fraction, galena and pyrite, ending in a variety of results ranging from ca. 2.0–1.6 Ga, with a concordant age of 1876 ± 9 Ma concluded to be the age of the diabase (Kouvo, pers. comm.).

The aim in this study is (i) to obtain paleomagnetic data from all Keuruu diabase dyke outcrops, (ii) to explain the asymmetry between normal and reversed data, and (iii) to obtain a late Paleoproterozoic key pole for Baltica, which includes a precise U-Pb age

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http://dx.doi.org/10.1016/j.precamres.2016.10.013
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and baked contact test. We further aim (iv) to discuss the implications of the pole in a late Paleoproterozoic paleogeographic reconstruction.

2. Geological background

The Keuruu diabase dyke swarm is situated within the Central Finland Granitoid Complex (CFGC) (Marmo, 1963; Marmo and Mikkola, 1964). The CFGC was formed 1.89–1.86 Ga ago during the peak phase of the Svecofennian orogeny, where the Svecofennian island arc system collided against the Archean craton Karelia (Nironen, 2003; Lahtinen et al., 2005). Granitoids are predominant rocks in the CFGC together with granodiorites, diorites and schists, as well as numerous gabbroic bodies. The U-Pb zircon age of the Keuruu granodiorite is 1.88 Ga (Huhma, 1986). The Keuruu diabase dyke swarm crosscuts these rocks (Nironen, 2003), as well as the Keuruu gabbro (Fig. 1).

The Keuruu dykes strike NW-SE (Fig. 1) and have near-vertical dips. Anisotropy of magnetic susceptibility results by Puranen et al. (1992) divide the Keuruu dykes into three groups based on magnetic fabric, namely horizontal (h), vertical (v) and mixed (m) orientations, which relates to the manner of magma flow of the dykes (Puranen et al., 1992). Puranen et al. (1992) suggested that fractionation of mafic magma in the upper part of the crust produced a low density (2600–2700 kg/m^3) tholeiitic magma. In a granodiorite surrounding with a density of ca. 2670 kg/m^3, the tholeiitic magma possibly reached its level of neutral buoyancy (LNB) near the present erosion level and intruded laterally along the LNB producing horizontal foliations and lineations parallel to the structural strike of the group h dykes. In a diorite or gabbro surrounding, with a slightly higher density of 2800–2950 kg/m^3, the LNB was located higher than the present erosion level resulting in group v dykes with weakly developed, gently plunging lineations and vertical foliations. Group m dykes possibly arose from buoyancy conditions between those of group h and v dykes.

Keuruu diabase dykes are chemically similar to each other, regardless of magnetic fabric, and they are also chemically similar to the gabbro of Keuruu (Fig. 2; Marmo and Mikkola, 1964; Puranen et al., 1992). Their low Ti and Zr contents at a given MgO are similar to those of some of the closest Svecofennian volcanic rocks, and largely distinct from generally more Ti- and Zr-enriched anorogenic mafic dyke swarms of southern Fennoscandia (Fig. 2). The coarser central parts of wider dykes are ophitic, subophitic or intergranular, containing plagioclase laths and hornblende grains (pseudomorphs after pyroxene) as the main minerals. The finer marginal parts are porphyritic in texture and contain fresh pyroxenes (orthopyroxene and augite) and some plagioclase phenocrysts (Puranen et al., 1992). The silicate matrix of the dykes includes biotite, chlorite, apatite, epidote, and titanite (Marmo and Mikkola, 1964; Puranen et al., 1992). A microscopic study of the Keuruu diabases revealed titanomagnetite as the predominant magnetic mineral (Puranen et al., 1992). The general lack of secondary mineral inclusions in these grains indicate their primary nature, although some secondary overgrowths and sporadic secondary grains (platy magnetite, hematite, pyrite, and pyrrhotite) have also been observed (Puranen et al., 1992). Thin section photographs of two of the samples of this study are presented in Fig. 3.

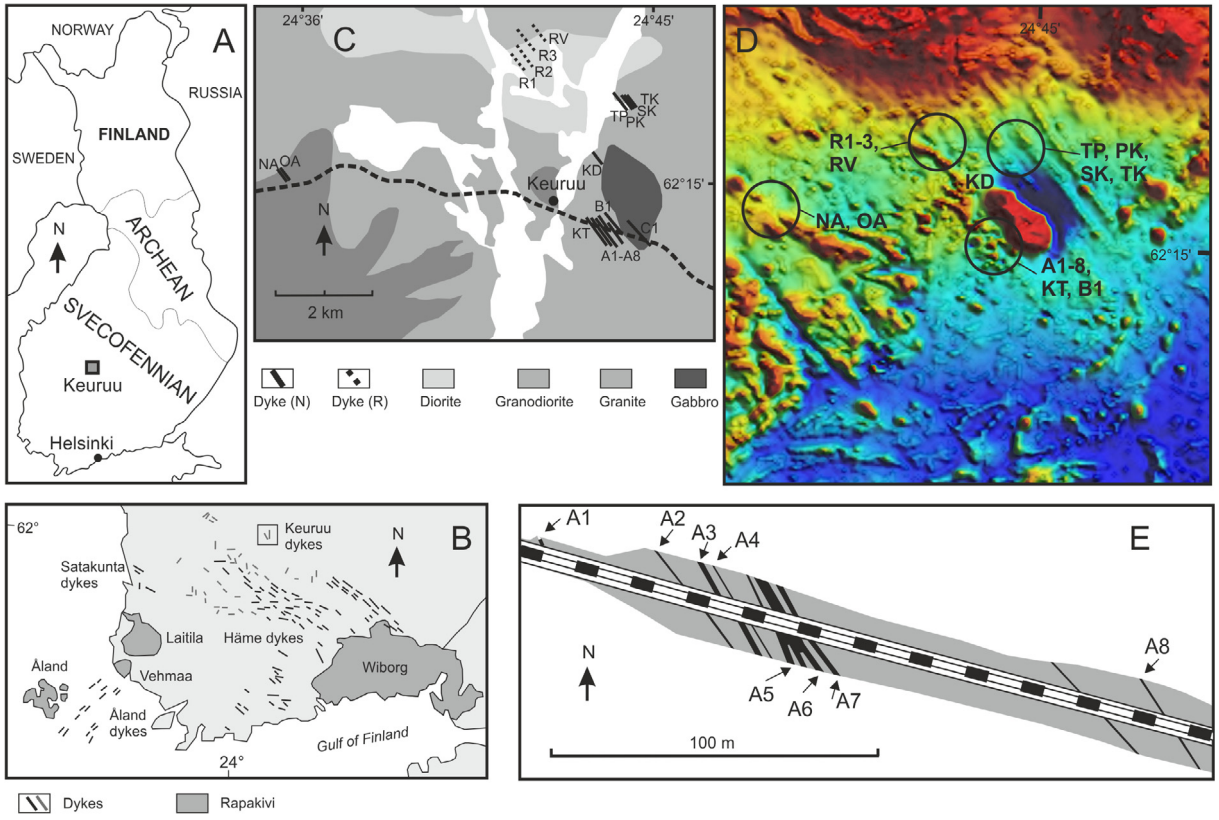


Fig. 1. (A) Simplified geological map of Finland (box show Keuruu study area). (B) Dyke map of Southern Finland (modified after Rämö and Haapala (2005)). (C) Geological map of the study area with the location of sampled dykes (modified from Puranen et al. (1992)). Normal (N) and reversed (R). (D) Aeromagnetic anomaly map of the study area (copyright: GTK airborne geophysical database). Nominal terrain clearance 30 m and line spacing 200 m. Flight line direction mainly N-S. Dyke outcrop sites encircled. SE-NW diagonal features in the bottom left corner of the aeromagnetic map corresponds with faults. (E) Close-up of the site of dykes A1-8, KT and B1 (the railway outcrop).

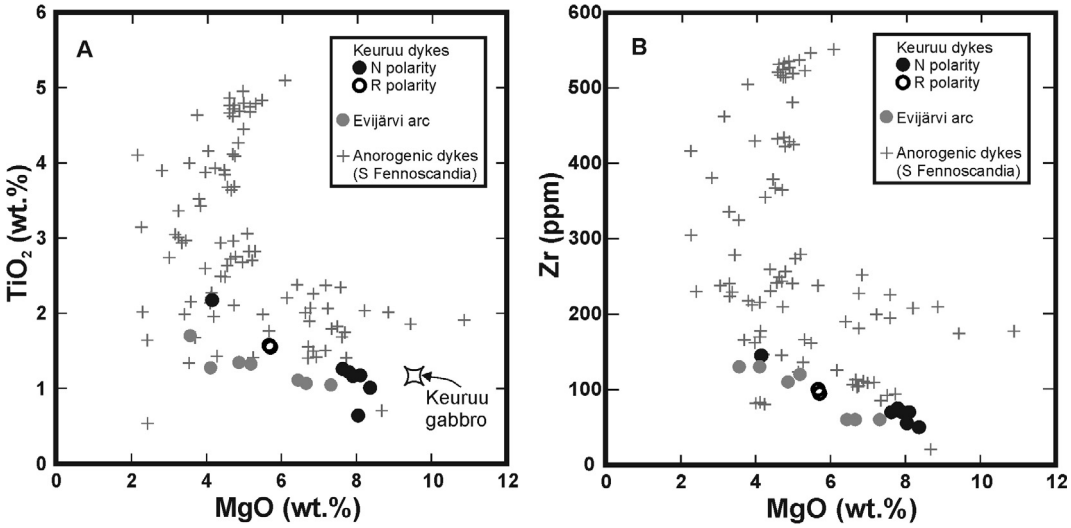


Fig. 2. Geochemistry of the Keuruu dykes presented in (A) MgO (wt%) vs. TiO_2 (wt%) and (B) Zr (ppm) vs. MgO (wt%) diagrams. The compositions of the Keuruu gabbro (no Zr data available; Puranen et al., 1992), ~1.9 Ga Evijärvi metavolcanic rocks (MORB/arc-type Patana and Isokylä formations; Vaarma and Kähkönen, 1994), and younger anorogenic magma suites of southern Fennoscandia (Rämö, 1990; Lindberg et al., 1991; Lobaev, 2005; Luttinen and Kosunen, 2006) are shown for comparison. See Luttinen and Kosunen (2006) for additional unpublished data sources and a more detailed classification of the anorogenic magma suites.

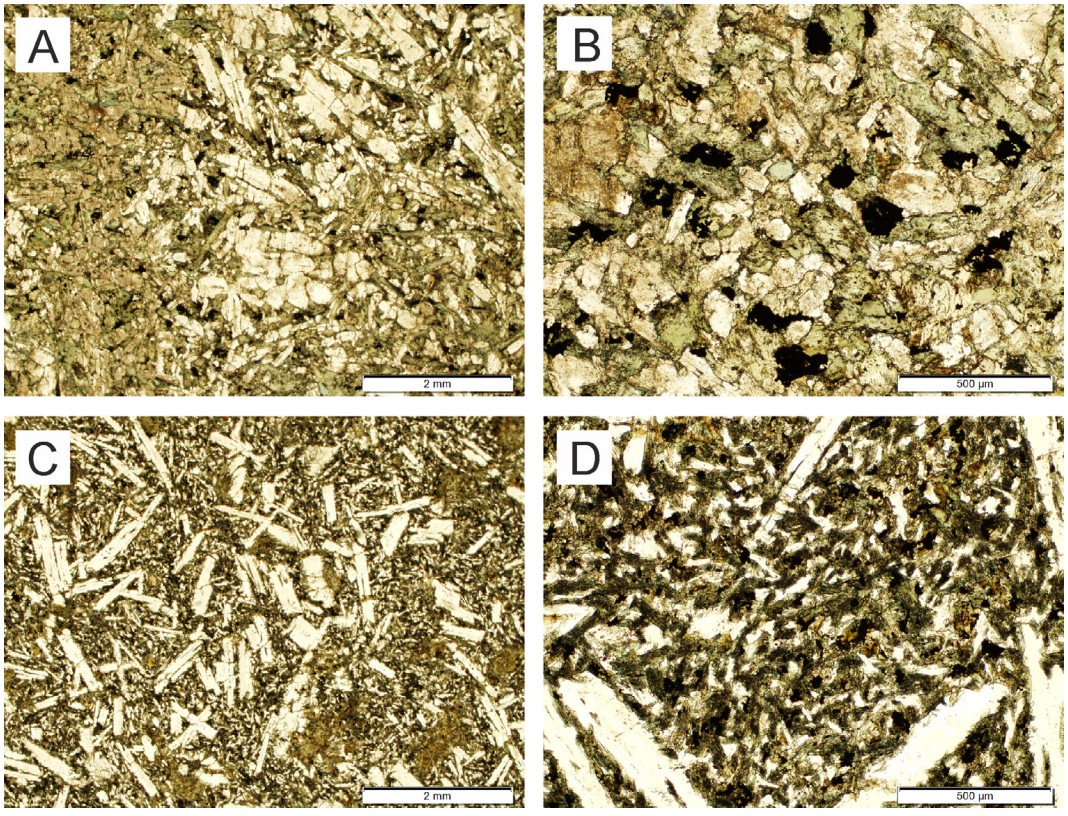


Fig. 3. Thin section photographs of the Keuruu dykes in polarized light. (A) N-polarity sample from dyke PK. Note the igneous texture and alteration of plagioclase on the left side of the image close to a fracture. (B) A closer view on the groundmass of the N-polarity sample from dyke PK. Note the slight alteration of plagioclase, greenish actinolite and chlorite, and the slightly overprinted euhedral to subhedral oxide crystals. (C) R-polarity sample from dyke R2. Note the igneous texture and freshness of the plagioclase crystals. Larger greenish spots are pseudomorphs of actinolite after pyroxene. (D) A closer view on the groundmass of the R-polarity sample from dyke R2. The groundmass consists mostly of biotite (brown), chlorite (green), and primary Fe-Ti oxides (opaque). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3. Methods

3.1. Sampling

Diabase dykes as well as baked and unbaked granodiorite, diorite, and gabbro host rock samples were collected from the Keuruu diabase dyke swarm for paleomagnetic measurements. Fig. 1 shows the location of the sampled dykes. Each dyke outcrop is treated as a separate dyke. Some of the samples were taken with a portable field drill, but the majority of the samples were taken as block samples. The samples were oriented with magnetic and sun compasses and the local magnetic declination was corrected. Multiple specimens were prepared from each sample; block samples were drilled in the laboratory and all the drill cores were sawed into cylinders (2.5 cm diameter (D), 2.2 cm length (h), with h/D = 0.9). Block samples taken from sites KD (0.3 m wide, N-polarity dyke) and R1 (1.2 m wide, R-polarity dyke) were also used for U-Pb geochronology measurements.

Block samples from dykes R1 (Saukko) and KD (Kelloniemi) (Fig. 1B) were prepared for Laser Ablation Multicollector Inductively Coupled Plasma Mass Spectrometry (LA-MC-ICPMS) U-Pb dating at the Geological Survey of Finland (GTK) in Espoo. These samples were assigned sample codes A2407 and A2408 respectively for the Finnish Rock Age Database of the GTK. In addition to these an existing sample in the GTK archives, A902, which was taken from an N-polarity dyke PK (Pirttivuori, Fig. 1B) in 1978, was prepared for U-Pb dating. Zircons were selected by hand-picking after heavy liquid and Frantz magnetic separation. The chosen grains were mounted in epoxy resin, and sectioned approximately in half, and polished accordingly. Back-scattered electron images (BSE) and cathode luminescence (CL) images were taken using scanning electron microscope (SEM) to target analysis sites on mineral grains.

3.2. Geochronology

U–Pb dating was performed using a Nu Plasma HR multicollector ICPMS at the Geological Survey of Finland in Espoo, Finland. A technique similar to Rosa et al. (2009) was used with the exception that an Analyte G2 193 nm laser microprobe was used. The analyses were made in static ablation mode with a beam diameter of 20 μm, a pulse frequency of 10 Hz, and a beam energy density of 2.07 J/cm^2. A single U–Pb measurement consisted of 30 s of on-mass background measurement, followed by 60 s of ablation with a stationary beam. Masses 204, 206 and 207 were measured in secondary electron multipliers, and mass 238 in the extra high mass Faraday collector. Ion counts were converted and reported as volts by the Nu Plasma time-resolved analysis software. ^{235}U was calculated from the signal at mass 238, based on ^{238}U/^{235}U = 137.88. Mass number 204 was used as a monitor for common ^{204}Pb. Raw data were corrected for background, laser induced elemental fractionation, mass discrimination, and drift in ion counter gains, and reduced to U–Pb isotope ratios by calibration to concordant reference zircons of known age, using protocols adapted from Andersen et al. (2004) and Jackson et al. (2004). Standard zircon GJ-01 (609 ± 1 Ma; Belousova et al., 2006), an in-house standard zircons A1772 (2711 ± 3 Ma/TIMS; 2712 ± 1 Ma/SIMS; Huhma et al., 2012) were used for calibration. In addition, sample A382 (1877 ± 2 Ma; Huhma et al., 2012) was measured as a zircon reference sample to check the calibration (1880 ± 5 Ma; MSWD of concordance = 2.7; n = 17). Age related common lead (Stacey and Kramers, 1975) correction was used when the analysis showed common lead contents above the detection limit. The calculations were done off-line, using an interactive spreadsheet program written in Microsoft Excel/VBA by Tom Andersen (Rosa et al., 2009). To compensate for drift in instrument sensitivity and Faraday vs. electron multiplier gain during an analytical session, a correlation of signal vs. time was assumed for the reference zircons. A description of the algorithms used is provided in Rosa et al. (2009).

Plotting of the U–Pb isotopic data and the age calculations were performed using the Isoplot/Ex 3 program (Ludwig, 2003). All the ages were calculated with 2σ errors and without decay constants errors. Data-point error ellipses are at the 2 σ level.

3.3. Paleomagnetic, rock magnetic and petrophysical measurements

Measurements were carried out at the Solid Earth Geophysics Laboratory of the University of Helsinki (UH), Finland (Pesonen, 2006) and at the Paleomagnetic Laboratory of the Geological Survey of Finland (GTK), Espoo (Pesonen, 1989).

Magnetic susceptibility, intensity of natural remanent magnetization (NRM), and density were measured for each specimen using the Risto-5 petrophysics assembly, and the Königsberger (Q) ratios were calculated. Furthermore, anisotropy and chemical composition results of Keuruu dykes obtained by Puranen et al. (1992) were replotted for normal and reversed polarity dykes to determine possible difference between the two polarities.

Stepwise alternating field (AF) demagnetizations were done using a single axis Schonstedt SD-1 demagnetizer up to 100 mT (GTK) and a three-axis demagnetizer with maximum field up to 160 mT, coupled with a 2G–DC SQUID magnetometer (UH). Stepwise thermal demagnetizations were performed using a Schonstedt TSD-1 furnace (GTK) and an ASC Model TD48-SC controlled atmosphere (Ar) furnace (UH). Remanent magnetizations were measured with a 2G–DC cryogenic magnetometer and a GTK-built spinner magnetometer. Both alternating-field (AF) and thermal demagnetization procedures were carried out. Vector components were visually identified using stereographic and orthogonal projections (Zijderveld, 1967) and the directions were calculated by a least squares method (Leino, 1991), with a maximum angular deviation (MAD) of ca. 6°. Occasionally vector endpoints of the demagnetization treatments were used to isolate components. Mean remanence directions for the different components were calculated according to Fisher (1953), giving a unit weight to each sample (each specimen has a unit weight within a sample) to compute site mean directions and corresponding paleomagnetic poles (Irving, 1964). Each dyke outcrop is treated as a separate site. The paleogeographical reconstructions were done with the GPlates program.

Magnetic carriers were studied by thermomagnetic analysis of selected specimens using an Agico CS3-KLY-3S Kappabridge system (UH), which measures the bulk susceptibility (k) of the samples during heating from −192° to 700 °C and cooling back to room temperature (in Argon gas). Curie temperatures were determined using the Cureval 8.0 program (http://www.agico.com). Magnetic saturation (J_s), remanence of saturation (J_r), coercivity (H_c), and remanence coercivity (H_{cr}) were measured using a MicroMag^{TM}3900 Vibrating Sample Magnetometer (VSM) by Princeton Measurement Corporation with a maximum field of 1T.

4. Results

4.1. U-Pb geochronology results

Mafic dyke sample A2407 Saukko (dyke R2, of reversed polarity) from the Keuruu swarm yielded only a few zircon grains. They are prismatic, rather large (>100 μm) showing either weak oscillatory zoning or a hazy texture in cathode luminescence (CL). Only five U-Pb analyses were made from this sample (Appendix A). The concordant age of the zircons is 1870 ± 9 Ma (Fig. 4A) and this sets the

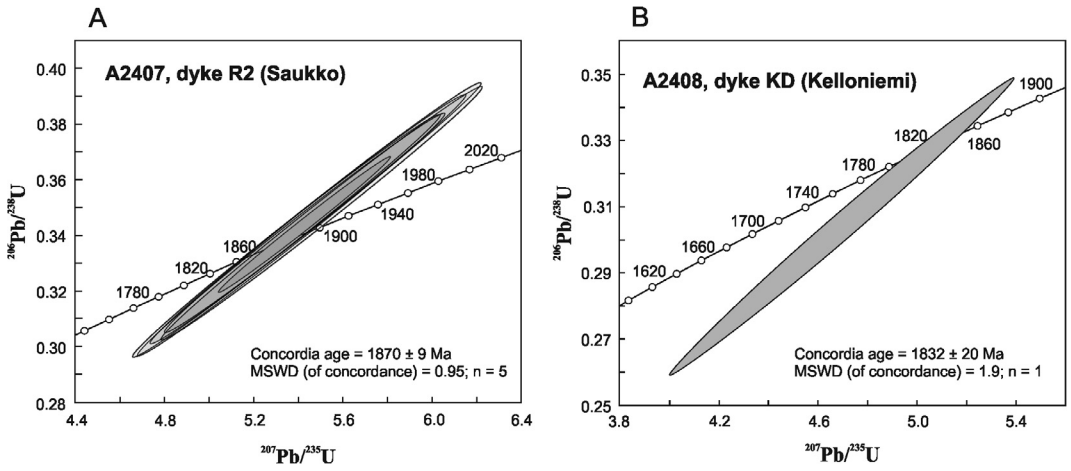


Fig. 4. Concordia plot for zircon LA-MC-ICPMS U-Pb data. (A) Sample A2407 Saukko (reversed polarity dyke R2), (B) Sample A2408 Kelloniemi (normal polarity dyke KD).

maximum crystallization age for the diabase. Due to the large size of the zircons and their euhedral normal morphology, we cannot rule out the possibility of an inherited origin of these zircons.

Sample 2408 Kelloniemi (dyke KD), an N polarity dyke from the Keuruu swarm, yielded only three zircon grains. Only one U-Pb analysis was made on a BSE-pale homogeneous and CL-pale, texturally hazy zircon domain. It is marginally concordant with a $^{207}Pb/^{206}Pb$ age of 1832 ± 20 Ma (Fig. 4B). However, only one result from this site renders the age inconclusive.

Sample A902 Pirttivuori (dyke PK, of normal polarity) from the Keuruu dyke swarm yielded plenty of zircons with varying grain-size, morphology (euhedral prismatic, rounded supposedly metamorphic), and internal texture (oscillatory zoned, dark to bright, some sector zoned, homogeneous, spotty in CL-images). All together 66 zircon domains were dated using LA-MC-ICPMS (Appendix A). The data are concordant (Fig. 5) and can be split into Archean (2.7–2.6 Ga) and Proterozoic age groups (Fig. 5). Archean zircons are not known from CFGC, but it is possible that this Archean material has been derived from unexposed parts of the local crust (e.g., from metasedimentary rocks or from hidden Archean crustal blocks; see Lahtinen, 2000; Andersson et al., 2011). Ignoring the single ~1.4 Ga age analysis, the Palaeoproterozoic Pb-Pb ages determine one main Svecofennian age peak at 1.88 Ga and a small 1.80 Ga peak.

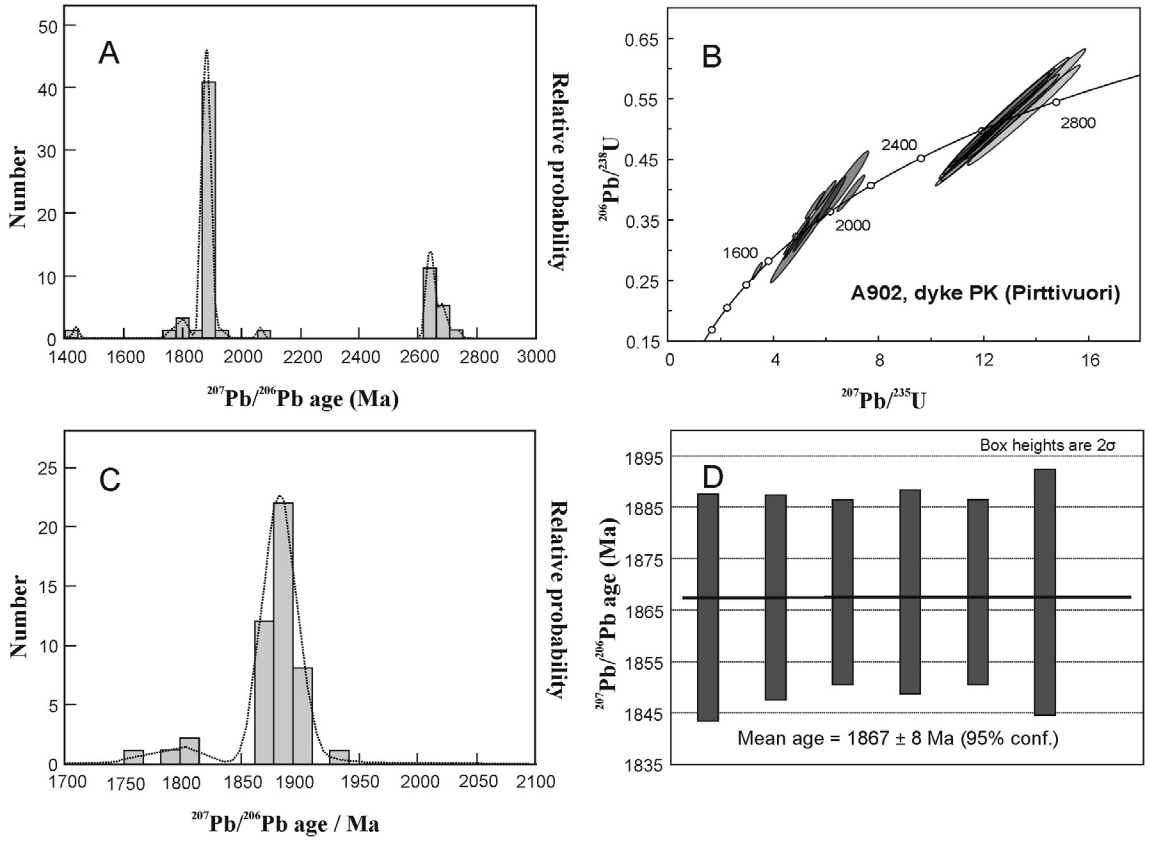


Fig. 5. (A) Probability plot for $^{207}Pb/^{206}Pb$ ages of zircon grains in sample A902 Pirttivuori (normal polarity dyke PK). (B) Concordia plot for zircon LA-MC-ICPMS U-Pb data, sample A902 Pirttivuoryi (dyke PK). (C) Probability plot for $^{207}Pb/^{206}Pb$ ages of Palaeoproterozoic zircon grains in sample A902 Pirttivuori mafic dyke. (D) Mean Pb-Pb age.

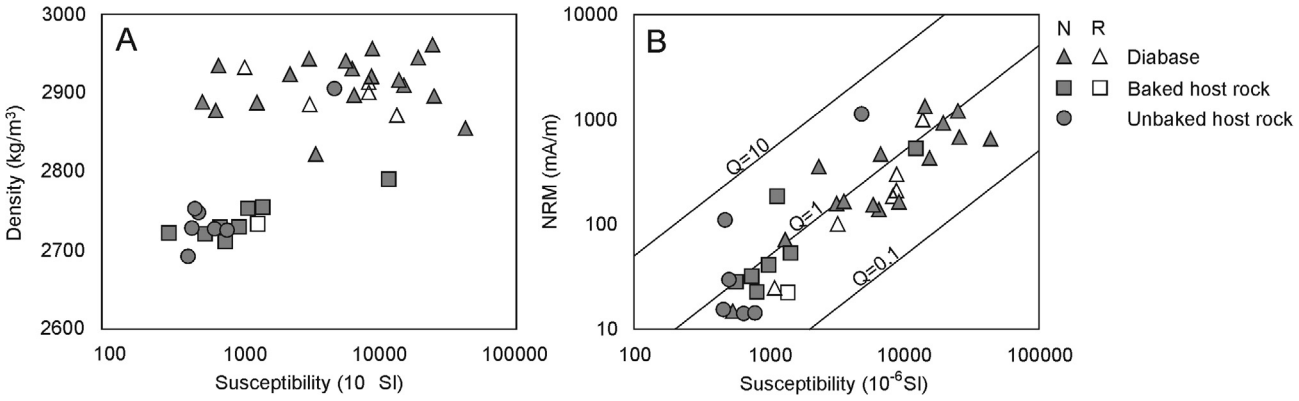


Fig. 6. (A) Density-susceptibility and (B) remanence-susceptibility diagrams for diabase and host rock samples of Keuruu. The line parameter is the Koenigberger (Q) ratio.

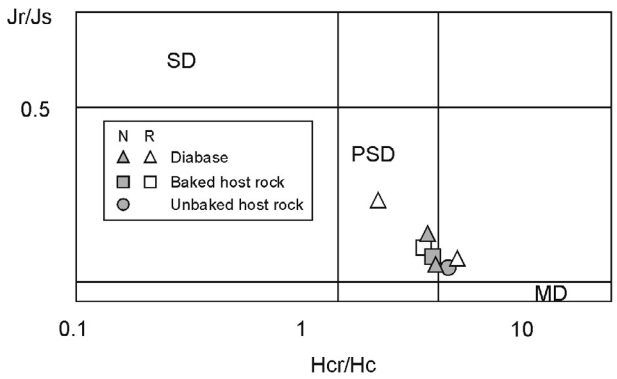


Fig. 7. Measured hysteresis ratios for Keuruu dyke and host rock samples, plotted on a Day plot (Day et al., 1977). J_r, saturation remanence; J_s, saturation magnetization; H_c, coercive force; H_{cr}, coercivity remanence; SD, single domain; PSD, pseudo single domain; MD, multi domain.

When speculating about the dyke ages the 1.80 Ga and the 1865–1870 Ma age groups should be considered, as the latter coincide with the formation of the host rock of the diabase dykes. It is evident that this rock contains inherited zircons. Six youngest Svecofennian zircon domains show a mean Pb-Pb age of ca. 1867 ± 8 Ma. Five of them were measured on CL/BSE homogeneous and CL-dark zircon domains, so these most probably indicate homogenization of the internal texture and thus also radiogenic lead loss. However, one of these measurements is from a large oscillatory zoned zircon, and therefore questions the possibility of the dyke age being determined by this age group. This is close to the Pb-Pb zircon age of 1876 ± 7 Ma obtained by Kouvo (pers. comm.) from the same dyke. The 1.8 Ga age group includes four analyses from only two zircon grains, and is therefore not considered: the zircons of this sample have been separated for bulk analyses several decades ago and some individual grains may represent sample contamination. There was no correlation between the ages and the domains of the zicon grains.

4.2. Petrophysical results

The mean petrophysical results for the Keuruu diabase dykes and host rock are illustrated in the susceptibility vs. density, and susceptibility vs. NRM plots in Fig. 6. There is a clear distinction between diabase (higher values) and host rock values, despite the large range of values and some overlap. Baking of the host rock seems to enhance both susceptibility and NRM in a similar fashion yielding similar or only slightly higher Q values for baked rocks. No difference between N and R results is seen in the petrophysical data. Similarly, no significance difference is shown by hysteresis values, as illustrated in Fig. 7, between diabase dyke and host rock samples, nor between N and R samples.

4.3. Paleomagnetic results

A total of 88 samples (202 specimens) from 22 dykes were measured, from which 69 samples (139 specimens) from 21 dykes gave stable results. The remaining samples were unstable, or had a maximum angular deviation (MAD)> 6° and were not included in the mean remanence directions (see Figs. 9 and 10 for examples of AF and thermal demagnetization behavior).

In general, diabase dyke samples show three components: (i) a steep downward component, resembling the Present Earth's field (PEF) at sampling site: D_{PEF} = 6°, I_{PEF} = 74°; (ii) a component with shallow to intermediate inclinations and northeasterly declinations (referred to as component B); and (iii) a dual polarity intermediate NNW downward or SSE upward component being the characteristic remanence component (ChRM). We employed the standard convention of assigning "normal" polarity to NW down directed ChRM vectors from Proterozoic rocks in Baltica. This is an arbitrary definition as the Precambrian geomagnetic polarity is uncertain due to the lack of continuity in the Baltica APWP prior to Phanerozoic.

4.3.1. Characteristic remananent magnetization (ChRM)

All dykes are either normally or reversely magnetized (Table 2). The normal polarity component was obtained from 53 samples (98

Table 1
Petrophysical properties of Keuruu diabase dykes and host rocks.

Rock type	B/N/n	D (kg/m$^{3)}$	K (10^{-6} SI)	J_0 (mA/m)	Q
Diabase dyke (N)	17/58/84	2911	10667	397	1.1
Diabase dyke (R)	5/24/77	2899	7047	325	1.0
Baked granodiorite (N)	7/13/57	2739	2268	113	1.3
Baked diorite (R)	1/3/8	2733	1369	23	0.4
Unbaked granodiorite	6/8/24	2754	1158	188	2.2

D = density; K = susceptibility; J_0 = intensity of natural remanent magnetization; Q = Königsberger's ratio. Magnetization polarity indicated by N or R after the rock type.

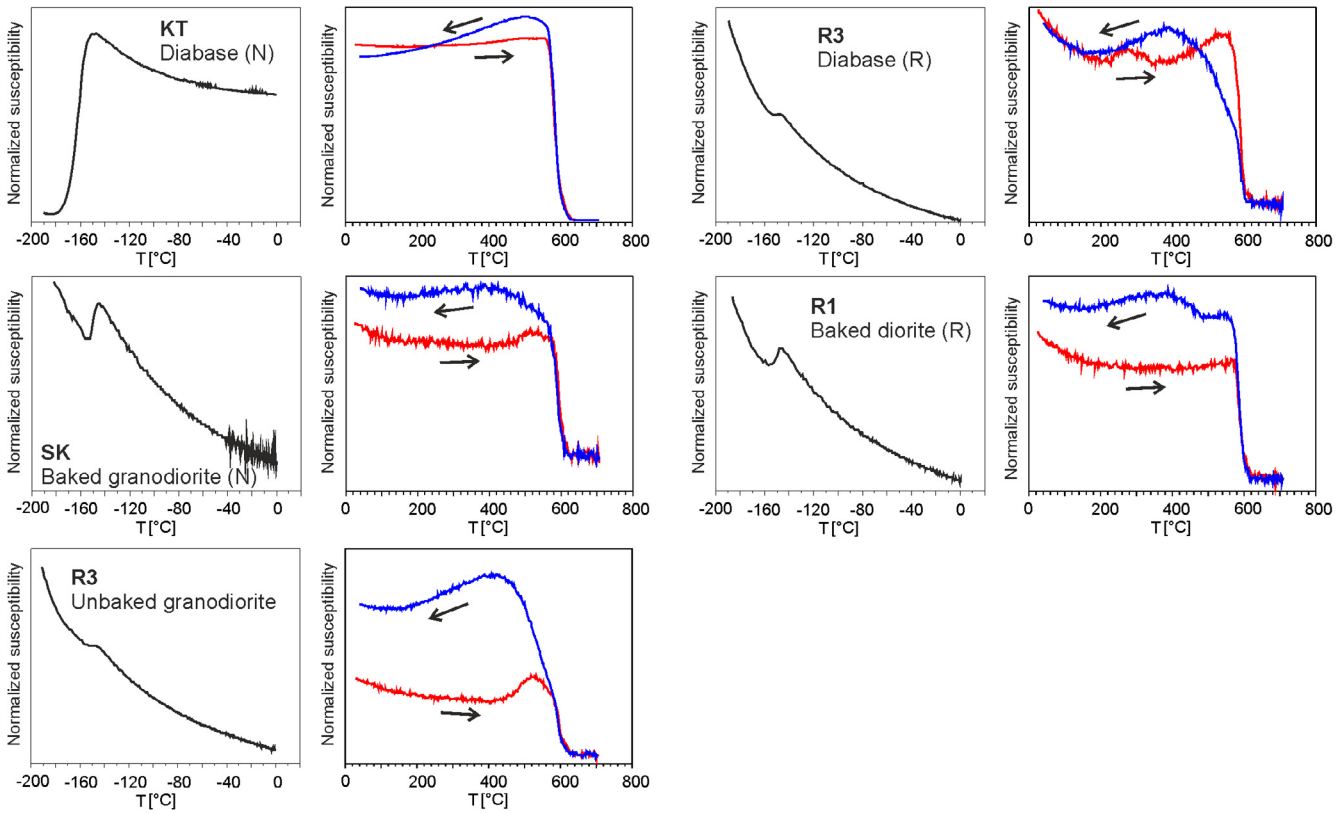


Fig. 8. Typical low-field thermomagnetic curves measured from −192° to 700 °C (from room temperature to 700 °C in argon) for selected Keuruu dykes and host rock samples. Susceptibility values were corrected with empty furnace values. Sample codes KT, SK, R1 and R3 correspond with sample codes in Fig. 1 and Table 2. Left plot shows low and right plot shows high temperature curve.

specimens) from 17 dykes, and reversed polarity component from 16 samples (41 specimens) from 4 dykes. The normal and reversed observed ChRM directions (Table 2) do not pass the McFadden and McElhinny (1990) reversal test, but reveal a distinct asymmetry of ca. 12° in inclination, and ca. 27° in declination. This confirms the results of Pesonen et al. (2012), showing asymmetry between 11 normal polarity, and 4 reversed polarity Keuruu dykes.

We also obtained stable paleomagnetic results from 27 host rock (granodiorite and diorite) samples (51 specimen) from 10 sites. We obtained baked samples from 7 normal polarity sites and 2 reversed polarity site, and unbaked host samples from a total of 6 sites. The normal and reversed directions obtained from the baked host rock (Table 2) correspond with the normal and reversed ChRM directions of the dykes. The unbaked host rocks (granodiorite and diorite) yielded a typical (N-polarity only) direction for Svecofennian rocks in Finland (Pesonen et al., 2003), and is indistinguishable from the normal polarity dyke and baked host rock directions (Fig. 11B)

Thermal (unblocking temperatures 540–600 °C) demagnetization indicate that the magnetic carrier of the ChRM is magnetite. To further confirm this, we conducted thermomagnetic measurements using susceptibility vs. temperature for both normally and reversely magnetized diabase dykes and host rock samples. The N and R polarity curves of diabases and baked granodiorites respectively, are nearly similar, but the unbaked host diorite (R3) is distinctly different. All curves reveal the presence of magnetite as the magnetic carriers. Sample R3 (Fig. 8) also shows a lower temperature phase (200–350 °C), probably due to pyrrhotite. We obtained Verwey transition peaks (Verwey, 1939) for all samples, indicating that magnetite is nearly stoichiometric.

4.3.2. Secondary components

It is difficult to isolate the steep secondary (PEF) and intermediate secondary (B) components, as they often co-exist in a similar coercivity range of ⩽20 mT or a blocking temperature range of ⩽300 °C (Figs. 7 and 8) in both dyke and host rock samples. The secondary component, however, is also carried at coercivities of >200 mT and high temperatures of >500 °C in some of the samples, where they co-exist with and contaminate the ChRM (Table 2). We interpret the low coercivity/blocking temperature component as a present Earth field (PEF) component, and the higher coercivity/blocking temperature component as a secondary component with ChRM contamination. In a great circle analysis by Pesonen et al. (2012), intersections of selected N and R specimens provide an estimate of D ≈ 47°, I ≈ 64° for the secondary component. Raiskila et al. (2011) obtained a similar direction (D = 42°, I = 64°) from the shocked or fractured Svecofennian target rocks (including granodiorite) of the Keurusselkä impact structure ca. 20 km South of Keuruu dykes, and interpreted it as an impact component acquired at ca. 1140 Ma based on ^{40}Ar-^{39}Ar dating on an impact generated pseudotachylitic breccia dyke (Schmieder et al., 2009). This may be one possible source of the secondary component, especially since recent evidence (Hasch et al., 2015) show that Keurusselkä impact event may have been much larger than previously thought. However an intermediate to steep NE component is observed widely in Baltica in earlier paleomagnetic studies of rocks ranging in age from ca. 2.44 Ga to 0.42 Ga (e.g. Mertanen et al., 1989; Preeden et al., 2009; Salminen et al., 2014; Klein et al., 2015). The paleomagnetic pole derived from this widespread component corresponds with Permian to Triassic (ca. 300–200 Ma) poles of Europe (Torsvik et al., 2012). Preeden

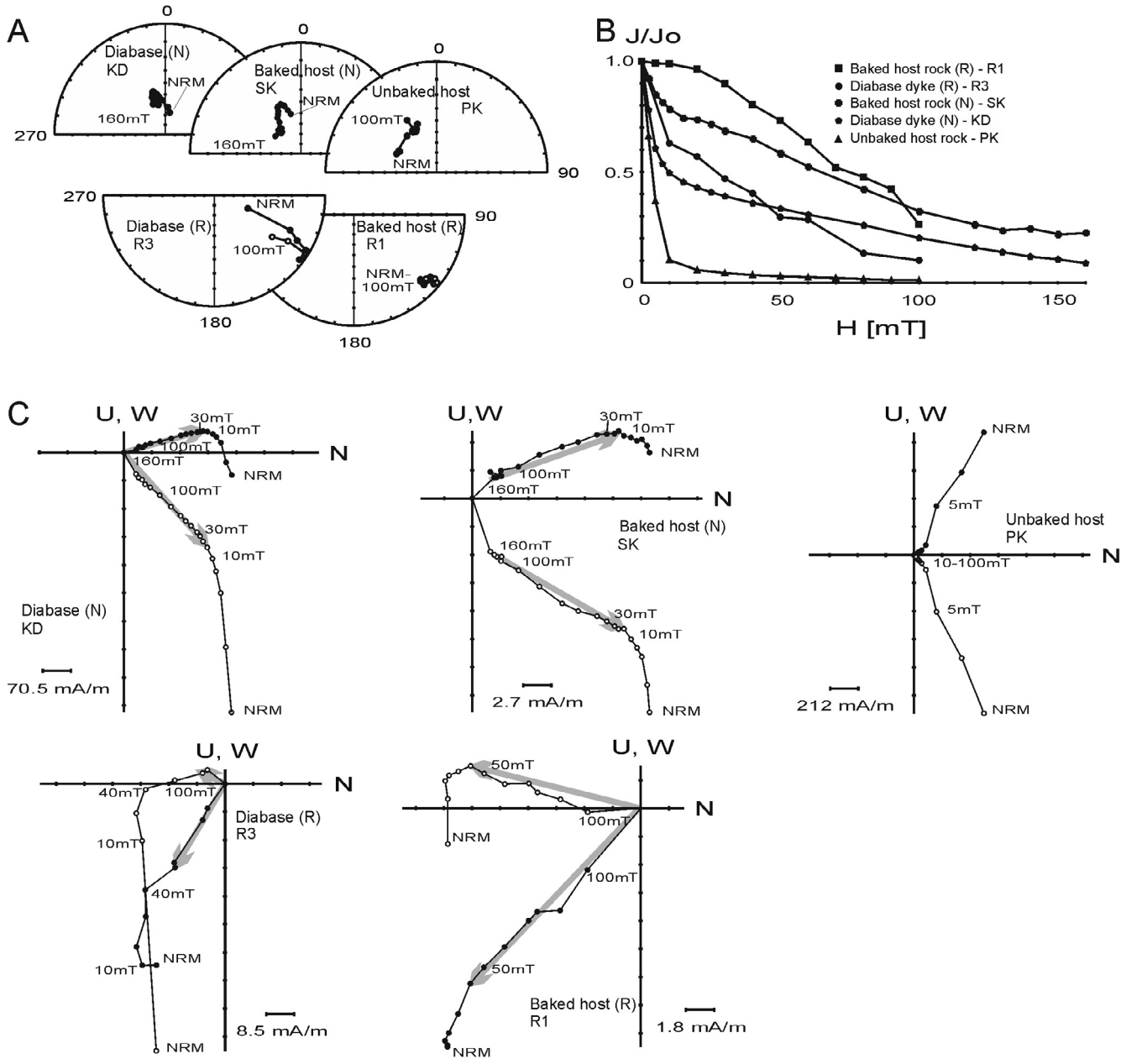


Fig. 9. Examples of AF demagnetization behavior. (A) Stereographic projections of directional data upon demagnetization. (B) Relative intensity decay. (C) Orthogonal (Zijderveld) demagnetization diagrams; open (closed) symbols denote vertical (horizontal) plane. N polarity host rock is granodiorite, R polarity host rock is diorite (Fig 1).

et al. (2009) interpret the cause of this component to be the migration of orogenic fluids related to the Hercynian orogeny at the southern margin of Baltica and/or the Uralian orogeny at the eastern margin.

4.3.3. Baked contact test

One of the basic requirements in paleomagnetic work is to verify that the observed magnetization direction is primary, recording the ambient field direction during emplacement (and subsequent cooling) of the rock formation, and not during later events. Since we deal with remanent magnetization of thermal origin in this study, it can be tested by the baked contact test (Everitt and Clegg, 1962). A positive baked contact test requires that the dyke and its baked host rock carry similar stable remanence directions, and that the unbaked host rock carries a different direction (e.g. Irving (1964), Buchan (2013)).

Baked-contact tests were performed at two R polarity dykes (R1 and R3) and one N polarity dyke (A4). The baked host rock were sampled less than 5 cm from the contact at each of the three dykes. At the N polarity dyke the unbaked host rock was sampled 8 m from the contact. One unbaked host rock sample is considered for both R polarity dykes, and was sampled ca. 1.5 m from the nearest (0.9 m wide) dyke. In the case of the R dykes, the dyke and the baked host rock have shallow to intermediate southeasterly (reversed polarity) upward pointing magnetization directions, while the unbaked Svecofennian host rock shows a typically northwesterly downward direction. For dyke R1 samples show a mean direction of D = 132.1° and I = −18.1° with α_{95} = 5.1° (6 samples, 14 specimen). The baked host rock (granodiorite) for this site shows a mean direction of D = 134.0° and, I = −24.5° (2 samples, 6 specimen). For dyke R3 samples show a mean direction of D = 141.3° and I = −24.7° with α_{95} = 16.8° (5 samples, 11 specimen).

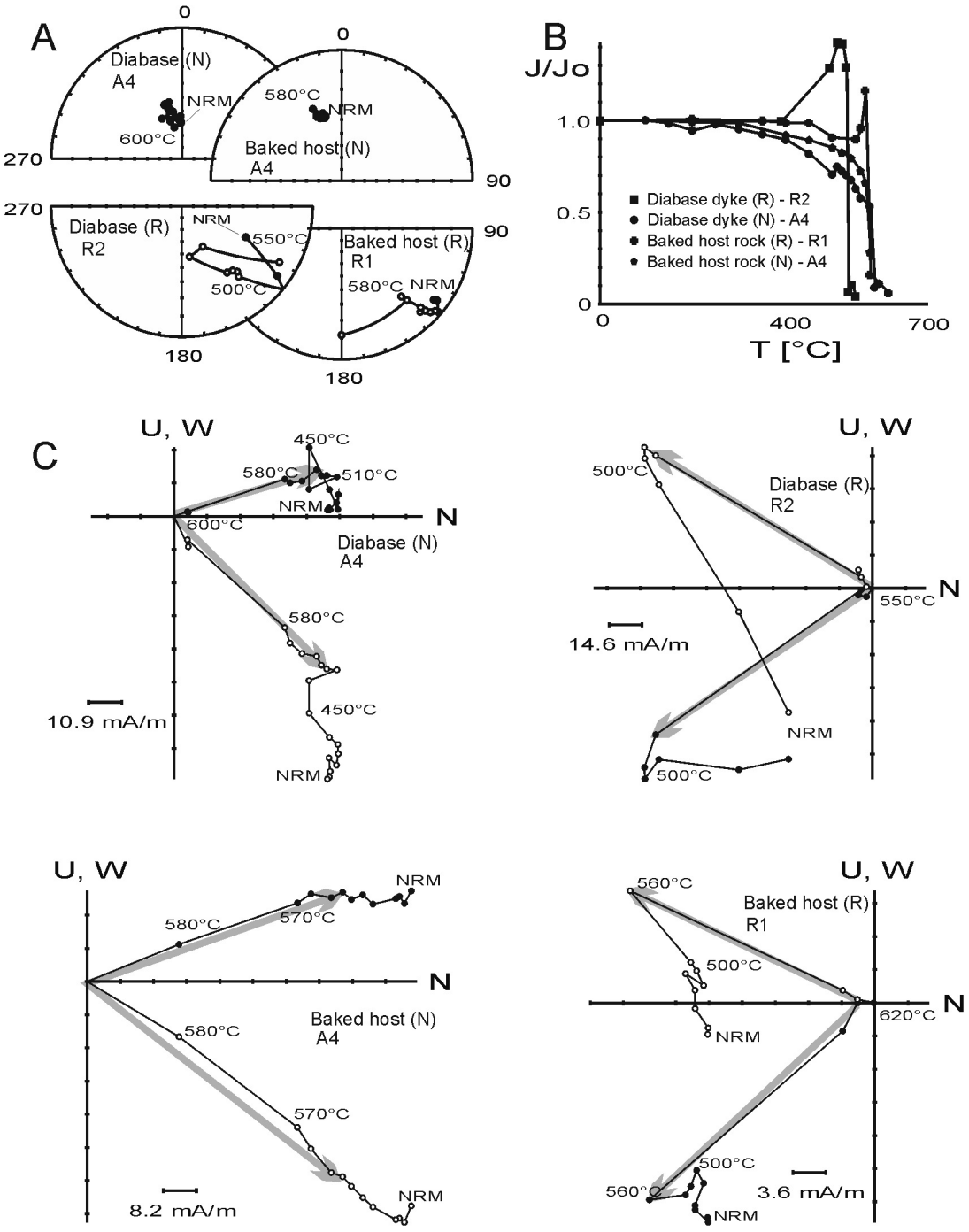


Fig. 10. Examples of thermal demagnetization. (A) Stereographic projections of directional data upon demagnetization. (B) Relative intensity decay. (C) Orthogonal (Zijderveld) demagnetization diagrams; open (closed) symbols denote vertical (horizontal) plane. N polarity host rock is granodiorite, R polarity host rock is diorite (Fig. 1).

The baked host for this site shows a mean direction of D = 139.2° and, I = −16.1° (2 samples; 4 specimen). Unbaked host for these dykes shows D = 308.0° and I = 18.0°, which is close to the typical Svecofennian N polarity direction. Although the unbaked ChRM for this site is based on one sample only, the unbaked host rock at other Keuruu sites (Table 2) also show a Svecofennian N polarity direction, as well as unshocked target rocks of the Keurusselkä impact structure ca. 20 km to the South (Raiskila et al., 2011). Based on these positive baked contact tests we interpret the R polarity component as the primary characteristic remanent magnetization.

A baked contact test for N polarity Keuruu dykes is difficult since the ChRM direction of the dykes is indistinguishable from the ChRM direction of the host rock (Fig. 11B), as expected due to their similar age. However, when looking at the NRM intensity decay behavior during the thermal demagnetization treatments for N polarity dykes (4 samples from 3 dykes) with associated baked and unbaked host rocks (dykes A1, A4, KT) the NRM intensity decay curves of the baked host rock (5 samples from 3 sites) are strikingly similar to the diabase decay curves (both are shoulder types typical for magnetite carrying TRM) and unlike the much softer decay curves for unbaked host rock (2 samples from 2 site).

Table 2
Palaeomagnetic results from Keuruu diabase their host rocks.

Site	Locality	Str (°)	Glat (°)	Glon (°)	W (m)	Gr	Pol	B/N/n	Dec (°)	Inc (°)	α_{95} (°)	k	Plat (°N)	Plon (°E)	A95 (°)	K
Diabase normal polarity																
KT	Railway		62.25	24.73			N	1/5	356.2	41.8			51.7	210.3		
A1	Railway	331	62.25	24.73	0.2	h	N	2/5	344.2	34.5			45.4	226.2		
A2	Railway	322	62.25	24.73	0.5	h	N	4*/8	336.5	37.2	9.2	101.5	45.7	236.7	7.4	156.4
A3	Railway	328	62.25	24.73	1.2	m	N	3*/5	344.3	22.4	22.1	32.1	38.2	224.3	22.2	32.0
A4	Railway	330	62.25	24.73	0.15	m	N	7*/11	344.4	40.7	7.9	59.3	49.7	227.3	8.7	49.3
A6	Railway	325	62.25	24.73	0.8	h	N	3*/5	339.5	22.7	18.9	43.4	37.7	230.0	13.9	79.5
A7	Railway	324	62.25	24.73	1.0	h	N	2/2	346.4	31.5			43.9	222.8		
A8	Railway	327	62.25	24.73	0.4	h	N	1/2	341.6	45.7			52.8	232.5		
B1	Railway	321	62.25	24.74	0.7	m	N	3*/6	341.6	44.2	14.5	73.1	52.1	232.3	11.4	117.8
C1	Railway	315	62.25	24.75	0.5	v	N	4*/8	341.9	32.6	15.4	25.7	44.3	228.3	11.8	43.0
NA	Haapamäentie	315	62.26	24.60			N	3*/6	351.1	23.4	34.4	13.9	40.0	216.6	29.6	18.4
OA	Haapamäentie	315	62.26	24.60	0.45		N	2/3	358.4	35.8			48.5	206.7		
PK	Pirttivuori	320	62.28	24.74	1.0	h	N	4*/6	353.3	28.1	12.2	57.5	42.5	213.3	12.7	53.4
SK	Pirttivuori	320	62.28	24.74	0.15		N	3*/4	348.2	39.5	19.9	39.5	49.8	221.7	18.0	48.0
TK	Pirttivuori	320	62.28	24.74	0.4		N	3*/4	354.4	45.0	10.9	130.1	54.2	213.5	12.0	107.4
KD	Kelloniementie	330	62.27	24.73	0.3		N	5*/7	334.7	46.6	5.1	223.4	52.0	242.6	5.4	203.0
TP	Kelloniemi	331	62.28	24.74	1.0		N	2/5	346.8	54.9			61.9	227.8		
Mean diabase normal polarity	***N***	***17*/53/98***	***346.1***	***37.0***	***4.8***	***57.2***	***47.9***	***224.2***	***4.0***	***79.8***						
Baked host rock (granodiorite) normal polarity																
KT	Railway		62.25	24.73			N	2/6	344.2	31.0			43.1	225.8		
A1	Railway		62.25	24.73			N	2/4	339.9	33.9			44.4	231.5		
A4	Railway		62.25	24.73			N	5*/9	339.3	41.3	10.9	50.1	49.5	233.9	9.8	61.6
SK	Pirttivuori		62.28	24.74			N	1/6	341.3	42.1			50.0	231.8		
TK	Pirttivuori		62.28	24.74			N	1/1	327.6	23.6			34.9	244.4		
TP	Kelloniemi		62.28	24.74			N	2/2	348.2	49.8			57.4	224.2		
KD	Kelloniementie		62.27	24.73			N	1/1	358.6	38.3			49.4	206.8		
Mean baked host normal polarity	***N***	***7*/14/29***	***342.3***	***37.5***	***8.6***	***50.2***	***47.4***	***229.1***	***8.0***	***58.1***						
Diabase reversed polarity																
R1	Saukko	322	62.28	24.70	1.2	v	R	6*/14	132.1	−18.1	5.1	175.2	−26.8	79.8	4.7	203.5
R2	Saukko	320	62.28	24.70	0.4	v	R	4*/13	136.9	−26.1	17.0	30.3	−33.2	77.4	11.7	62.7
R3	Saukko	320	62.28	24.70	0.9	v	R	5*/11	141.3	−24.7	16.8	21.6	−34.2	71.1	15.8	24.4
RV	Saukko		62.28	24.70			R	1/3	145.5	−30.3			−38.0	68.4		
Mean diabase reversed polarity	***R***	***4*/16/41***	***138.8***	***−24.9***	***8.3***	***123.8***	***−32.8***	***74.3***	***7.2***	***164.3***						
Baked host rocks (diorite) reversed polarity																
R1	Saukko		62.28	24.70			R	2/6	134.0	−24.5			−30.8	79.4		
R3	Saukko		62.28	24.70			R	2/4	139.2	−16.1			−28.4	72.1		
Mean baked host reversed polarity	**R**	**2/4*/10**	**136.7**	**−20.3**	**11.3**	**67.5**	**−29.6**	**75.7**	**9.7**	**91.6**						
Unbaked host rock (granodiorite and diorite)																
A1	Railway		62.25	24.73			N	2/4	353.1	44.6			53.8	215.2		
A4	Railway		62.25	24.73			N	2/2	330.1	46.7			50.6	248.0		
R3	Saukko		62.28	24.70			N	1/1	308.0	18.0			25.1	263.9		
SK	Pirttivuori		62.28	24.74			N	1/1	335.7	32.4			42.2	236.7		
PK	Pirttivuori		62.28	24.74			N	1/1	335.0	42.8			49.0	240.5		
KD	Kelloniementie		62.27	24.73			N	2/3	344.4	51.5			58.5	230.6		
Mean unbaked host normal polarity	***N***	***6*/9/12***	***333.0***	***40.3***	***14.6***	***22.1***	***47.6***	***241.3***	***13.8***	***24.5***						
Diabase dyke GRAND MEAN (combined N and R)	***C***	***21*/69/139***	***340.4***	***35.2***	***5.8***	***31.3***	***45.7***	***230.9***	***5.5***	***34.7***						
Mean secondary component (<20 mT. <300 °C)		***19*/47/71***	***24.8***	***68.4***	***8.0***	***18.5***	***75.7***	***131.1***	***11.5***	***9.5***						
Mean secondary component (>30 mT. >500 °C)		***18*/42/57***	***9.1***	***51.6***	***8.7***	***16.6***	***61.9***	***188.1***	***8.5***	***17.6***						

Site – corresponds with sample codes in Fig. 1. Str – Structural orientation (strike) of dykes. Lat/Lon – sampling latitude and longitude. Gr. – magnetic fabric group: horizontal (h). vertical (v) or mixed (m) foliations (Puranen et al., 1992). Pol – polarity of the isolated direction: N(R) – normal (reversed) polarity. B/N/n – number of analyzed sites/samples/specimens. * – denotes the number used to calculate mean value. Dec – declination. Inc – inclination. $\alpha 95$ – the radius of the 95% confidence cone in Fisher (1953) statistics. k – Fisher (1953) precision parameter. Plat/Plon – latitude/longitude of the pole. A95 – radius of the 95% confidence cone of the virtual geomagnetic pole (VGP). K – Fisher (1953) precision parameter of pole. A4 dyke and baked host rock means. And A1 unbaked host rock mean. Include samples from Palmu (1982).

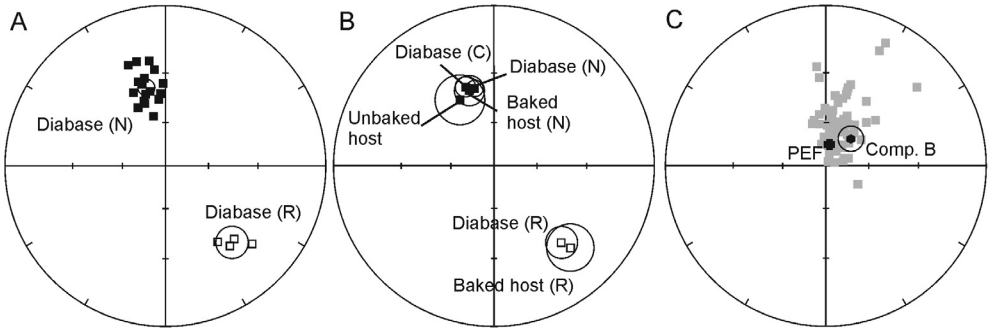


Fig. 11. (A) Stereoplot showing the site mean directions of the ChRM component of the diabase dykes. (B) Mean directions of diabase dykes, baked host rock and unbaked host rock. N, R and C indicate normal, reversed and combined polarity, respectively. Solid (open) symbols represent lower (upper) hemisphere vectors. (C) Sample means of secondary components from Keuruu dykes and host rocks shown with the present Earth field component (PEF), and component B (Pesonen et al., 2012).

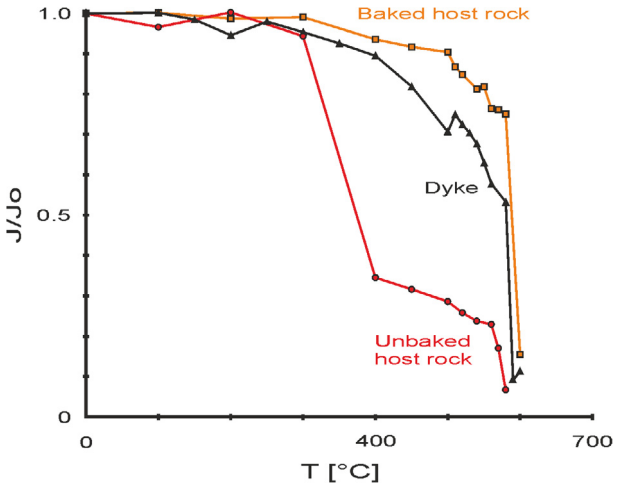


Fig. 12. NRM intensity decay with increasing temperature for dyke A4, and associated baked and unbaked host rock.

An example of the decay curves from site A4 is illustrated in Fig. 12. We therefore interpret the ChRM direction of the baked host rock to be derived from baking, suggesting a positive baked contact test also for N polarity dykes. In summary we propose that the Keuruu N and R polarity dykes carry a primary TRM or TCRM, recording the time of dyke cooling at ca. 1.87 Ga.

5. Discussion

5.1. Petrological considerations

On the basis of geochronological data alone, it would be possible to argue that the zircons are inherited and only give maximum ages for the dykes and that they could well be related to later anorogenic magmatic phases of southern Fennoscandia (major peaks at ~1.7–1.5 Ga and at ~1.25 Ga; e.g. Suominen (1991), Vaasjoki et al. (1991)). As already mentioned, however, the geochemical characteristics of the dykes correspond more closely to those of Keuruu gabbro and Svecofennian arc-related rocks from nearby metavolcanic belts than to those of the generally Ti- and Zr-rich anorogenic dyke swarms (Fig. 2). In addition, Marmo and Mikkola (1964) and Puranen et al. (1992) have described orthopyroxene from the groundmass of some the dykes. Its presence is likely related to relatively high pressures of crystallization (see Feig et al., 2006). In contrast, orthopyroxene is only found as occasional crystals and within autoliths in the anorogenic dykes of southern Finland (see, e.g., compilation by Aro and Laitakari (1987)). At ~1.8 Ga ago, the present erosional level lay at a depth of about 15 km, whereas the exposed rapakivi granites and related mafic dykes intruded the upper crust at a depth of ~5 km (Vaasjoki et al., 2005). Overall, these petrological observations support post-orogenic, not anorogenic, origin for the Keuruu dykes. Their intrusion in NW-SE trending fractures may well be related to post-collisional extension suggested to have taken place at ~1.87–1.86 Ga, after the peak of the Svecofennian orogeny at ~1.89 Ga (Nikkilä et al., 2015).

5.2. A Svecofennian key pole for Baltica

5.2.1. Primary magnetization

As the ChRM component of the Keuruu diabase dykes is carried by magnetite (confirmed by thermo-magnetic and thermal demagnetization data), and is supported by baked contact tests, we interpret it as a primary paleomagnetic magnetic component. The primary nature of the direction of magnetization can also be estimated on the basis of petrographical studies. Puranen et al. (1992) already estimated this to some degree, and concluded that although there was some evidence of secondary crystallization of magnetic minerals, most of the grains were primary titanomagnetite that crystallized from the basaltic magma. Our observations from two thin sections (N-polarity sample from dyke PK and R-polarity sample from dyke R2) confirm these conclusions: oxides are settled within the igneous fabric of the dykes and show minor evidence of secondary overgrowth as irregular branches and minor inclusions (Fig. 3). Overall, the samples show evidence of well-preserved igneous textures.

Unlike Puranen et al. (1992), we did not observe epidote as a minor phase in the thin sections. In fact, plagioclase was very pristine, although some secondary alteration (saussurization?) was observed close to fractures crosscutting the sample from dyke PK (Fig. 3). The only observed evidence of metamorphism or hydrothermal alteration are very fine-grained chlorite (+ minor biotite) in the groundmass and the pseudomorphs of actinolite after pyroxene (Fig. 3). Such paragenesis has also been described from ocean-floor sheeted dyke systems, in which it is thought to indicate that the alteration took place at temperatures of 350–380 °C at maximum (e.g. Alt et al. (1996)). Taking this into account, it is unlikely that the Keuruu dykes experienced metamorphic conditions above lower greenschist facies so that the unblocking temperatures of magnetite (540–600 °C) would have been reached.

As a side note, the N-polarity dyke sample was more altered than the R-polarity sample (the N-polarity group also exhibit higher loss-on-ignition values than the R-polarity group on average; Puranen et al., 1992), but whether this observation has some statistical and thus geological significance cannot be concluded.

5.2.2. Asymmetry

The normal and reversed polarity ChRM components do not pass the McFadden and McElhinny (1990) reversal test; they are not antiparallel in 95% confidence level. Non-antiparallel directions have already been obtained in Subjotnian (ca. 1.6 Ga) paleomagnetic studies from Fennoscandia (e.g. Pesonen and Neuvonen (1981), Salminen et al. (2014)). Several explanations for the asymmetry can be offered (see Pesonen and Neuvonen (1981)), including: (1) Anisotropies (either fabric or refraction), which can deflect the ChRM from the ambient field directions, (2) paleosecular variation (Buchan and Halls, 1990), (3) an age difference between R- and N-polarity intrusions with continental drift (change in latitude; Swanson-Hysell et al., 2014); (4) an unremoved secondary component (Palmer, 1970); (5) relative crustal tilting between R- and N-polarity "dominated" blocks (Halls and Shaw (1988)) and; (6) a dipole field that is contaminated by a permanent non-dipole field component (e.g. Pesonen and Nevanlinna (1981), Veikkolainen et al. (2014a)).

Similar ages were obtained for the normal and reversed polarity dykes of Keuruu, therefore the age difference and thus the APWP during the reversal crossing in time, is not a contributing factor in the non-antiparallel directions. The fabric anisotropy cannot cause the asymmetry; measured anisotropies were relatively small ($\leqslant$10%) with no difference between N and R fabric data (Puranen et al., 1992). Another anisotropy, which can deflect the direction of remanence from the ambient field, i.e. magnetic refraction (Strangway, 1961). However, the susceptibilities are relatively weak for the diabase dykes ($\leqslant 12{,}000 \times 10^{-6}$ SI) and similar in N and R dykes, and can therefore not significantly deflect the ChRM from the ambient field due to self-demagnetization by the dyke wall. We also consider relative crustal tilting between R- and N-polarity blocks unlikely as the Keuruu diabase dyke swarm occurs within a relatively narrow zone (Fig. 1.) with no significant crustal block boundaries between N and R blocks.

Asymmetry caused by a non-dipole field is difficult to prove since it would require a global analysis of dual polarity paleomagnetic data of 1.87 Ga formations. In the case of the Keuruu asymmetry, the possible non-dipole field must be non-zonal since the asymmetry is in both declination and inclination (Table 2). Furthermore, recent analyses of the Precambrian field suggest a GAD field more than previously thought (Veikkolainen et al., 2014a,b). Based on these data, we do not consider a permanent non-dipole field a cause of polarity asymmetry in Keuruu dykes.

We interpret the most likely cause for the asymmetry between the N and R polarity Keuruu diabase dykes, as well as the non-Fisherian distribution of the N directions (Fig. 10A), to be an unremoved secondary component. This is apparent from the demagnetization data of several specimens (e.g. KD in Fig. 8). We tested this through vector analysis, subtracting the secondary component (Table 1) from the N and R ChRM components, respectively. A component B intensity of 40% of the total intensity resulted in antiparallel N and R directions within the 95% confidence of the ChRM component.

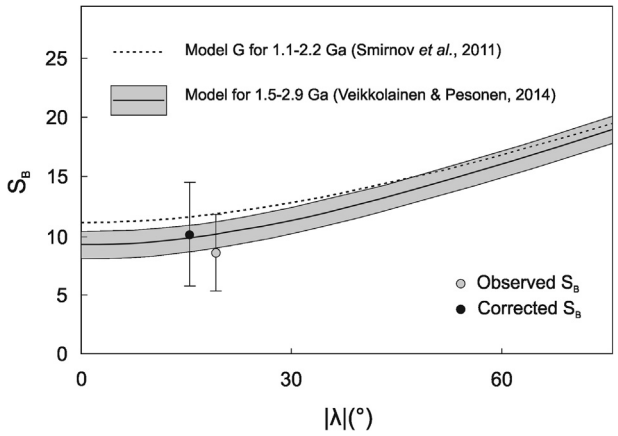


Fig. 13. The observed and corrected (for unremoved secondary component) dispersion of virtual geomagnetic poles (S_B) by the absolute value of paleolatitude for within site corrected paleomagnetic data for 11 normal polarity Keuruu diabase dykes.

As the reversed direction is carried by only 4 dykes, the paleosecular variation is not likely to be averaged out. The between site dispersion for 11 normal polarity dykes was calculated, after the within site dispersion was removed, and plotted with the 1.5–2.9 Ga PSV model (Veikkolainen and Pesonen, 2014) as well as with the 1.1–2.2 Ga "G" model of Smirnov et al. (2011; Fig. 13). The between site dispersion (S_B) of 8.5° at a paleolatitude of 19°, plots just below both models, however the error bars of the Keuruu result (±3.2°) overlaps with the PSV model by Veikkolainen and Pesonen (2014). When we apply a correction for the unremoved secondary component the between site dispersion ($S_{B)}$ of 10.2° ± 4.3° at a paleolatitude of 15° plots on the curve of the model by Veikkolainen and Pesonen (2014), with the error bars crossing the model by Smirnov et al. (2011). Secular variation is therefore averaged for the normal polarity distribution.

5.3. The assembly of Nuna

The 1.87 Ga paleogeographic reconstruction presented here (Fig 14), based on the Keuruu diabase dyke pole and coeval poles presented in Table 3, indicates the initial stages of formation of the supercontinent Nuna, which was assembled by at least ca. 1650–1580 Ma (Pisarevsky et al., 2014).

At 1.87 Ga, Baltica and Laurentia were not yet assembled. Sarmatia and Volgo-Uralia collided with the Fennoscandian Shield only between 1.82 and 1.80 Ga along the Russian collisional belt to form Baltica (Bogdanova et al., 2008, 2015; Elming et al., 2010). The trans-Hudson orogeny, connecting Superior-Nain and Slave-Rae-Hearne blocks of Laurentia, occurred between 1.91 and 1.81 Ga (Hoffman, 1988; Mitchell et al., 2014). Fennoscandia is positioned at low latitudes and aligns with the Superior-Nain block in a way that would proceed into the 1.8–1.2 Ga NENA configuration, which had been established by 1.75 Ga (Salminen et al., 2014)). The low to intermediate latitudinal positions of Baltica and Laurentia blocks are consistent with occurrences of evaporites and phosphates in both continents (Pesonen et al., 2003, and references therein). Overlapping paleomagnetic poles between 1880 Ma and 1786 Ma (Fig. 13) suggest minimal movement of Baltica during this period. The Superior poles for this period, on the other hand, are discordant. This does not necessary imply a rapid rotation of the Superior Craton, but is likely related to the relative movement between the eastern and western parts of Superior that occurred between 2070 and 1870 Ma (Evans and Halls, 2010).

Siberia possibly formed part of the Nuna core (Wingate et al., 2009; Evans and Mitchell, 2011). The 1878 Ma Lower Akitkanchaya and 1863 Ma Upper Akitkanchaya poles indicate a low latitudinal position of Siberia (Didenko et al., 2009). In the 1.87 Ga reconstruction we suggest a close proximity between Siberia and Slave. Several authors (e.g. Eglington et al. (2013), Ernst et al. (2016)) provide geological support for a connection between Southern Siberia and Northern Laurentia since ca. 1.9 Ga, however this connection is not supported by the existing paleomagnetic poles, as there are no coeval pole pairs between Slave and Siberia. The Slave APWP (Table 3, Fig. 14) consists of poles at ca. 1885 Ma, 1882 Ma and 1870 Ma (Mitchell et al., 2010), with a brief excursion at ca. 1875 Ma (Bingham and Evans, 1976). The Siberia APWP consists of the Lower and Upper Akitkanchaya poles (1878 Ma and 1863 Ma respectively) with no record of an excursion.

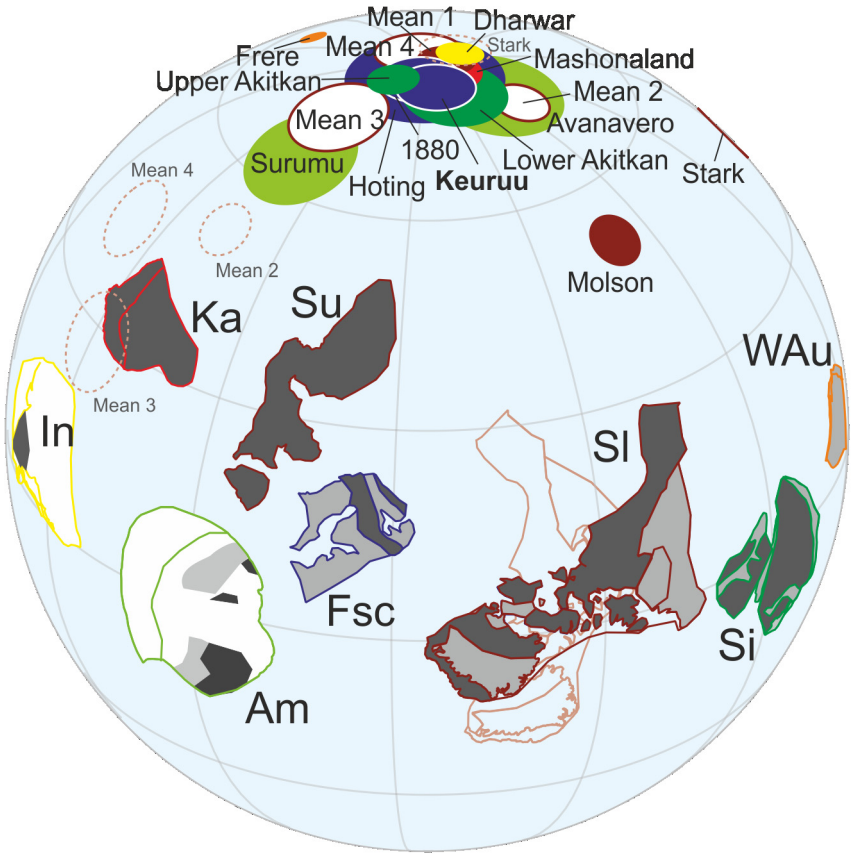


Fig. 14. Paleogeographic reconstruction of continents at 1.87 Ga based on the Keuruu dyke pole and coeval poles listed in Table 3. FSc = Fennoscandian Shield, Su = Superior Craton, Sl = Slave Craton, Am = Amazonia, Si = Siberia, WAU = Western Australia, In = India, Ka = Kalahari. The colour of the poles correspond with the colour of craton outlines. Closed and open Laurentia poles indicate Superior and Slave poles, respectively.

Table 3
Paleomagnetic poles used in the paleogeographic reconstruction at ca. 1.87 Ga (Fig. 14).

Formation	Plat (°N)	Plon (°E)	A95 (°)	Age (Ma)	Q(1–7)	References
Baltica						
Mean Svecofennian	44	234	5.7	1880	1111001 = 5	Elming and Pesonen (2010)
Keuruu diabase dykes	46	231	5.5	1870	1101111 = 6	This study
Hoting gabbro	43	233	10.9	1786 ± 10	1111001 = 5	Elming et al. (2009)
Laurentia						
Molson dykes B + C2	28.9	218	3.8	1877 + 7/−4, 1884 ± 2	1111111 = 7	Halls and Heaman (2000) Zhai et al. (1994) Evans and Halls (2010)
MEAN 1: Haig/Flaherty/Sutton	1	246	3.9	1870 ± 1	1111111 = 7	Elming and Pesonen (2010)
MEAN 2: Seton/Akaitcho/Mara	−6	260	4.0	1885 ± 5	1111111 = 7	Mitchell et al. (2010)
MEAN 3: Kahochella/Peacock Hills	−12	285	7.0	1882 ± 4	1100111 = 5	Mitchell et al. (2010)
MEAN 4: Pearson A /Peninsular sill/ Kilohigok basin sill	−22	269	6.0	1870 ± 4	1101111 = 6	Mitchell et al. (2010)
Stark formation	−15	215	4.9	1876 ± 10	1110111 = 6	Bingham and Evans (1976)
Amazonia						
Surumu volcanics	27	235	8.1	1970 ± 10	1111101 = 6	Bispo-Santos et al., 2014a
Avanavero mafic rocks	48	208	9.2	1789 ± 3	1111101 = 6	Bispo-Santos et al., 2014b
Siberia						
Lower Akitkan Khibelen group (Malaya Kosa formation)	−31	99	3.5	1878 ± 4	1111101 = 6	Didenko et al. (2009)
Upper Akitkan group	−22	98	5.2	1863 ± 9	1110101 = 6	Didenko et al. (2009)
W. Australia						
Frere formation	45	40	1.8	1890 ± 10	1110111 = 6	Williams et al. (2004) Rasmussen et al. (2012)
India						
Dharwar 1.88 Ga dykes	37	334	3.2	1885 ± 3	1111110 = 6	Belica et al. (2014)
Kalahari						
Mashonaland dolerites	−8	158	5.0	1871.9 ± 2.2, 1881.1 + 5.9/−5.1, 1882.7 + 1.6/−1.5	1111111 = 7	Bates and Jones (1996) Hanson et al. (2011) Söderlund et al. (2010)

Plat, Plon is latitude and longitude of the paleomagnetic pole. A95 = 95% confidence circle of the pole, $Q_{(1-7)}$ = Van der Voo reliability criteria of paleomagnetic poles.

Although there are no high quality paleomagnetic poles for Amazonia at ca. 1.87 Ga an interpolation was used (Table 3), positioning Amazonia at near equatorial positions from where it could proceed into a SAMBA configuration (Johansson, 2009) with Baltica. The Western Australian craton is situated at low latitudes in the 1870 Ma reconstruction (Fig. 14) from where it possibly moved northwards to form the SWEAT-like configuration with Laurentia during the Mesoproterozoic.

The 1.88 Ga Dharwar dykes pole positions India at an equatorial position. Based on the 1.77–1.27 reconstructions of Pesonen et al. (2003), Jacobs et al. (2008), and Pisarevsky et al. (2014) in which Kalahari is surrounded by ocean, Kalahari is positioned as a "lone" continent in the Northern hemisphere in the 1.87 Ga reconstruction (Fig. 14). Based on a mismatch of the 1.87–1.88 Ga Mashonaland pole from the Zimbabwe craton (Bates and Jones, 1996) and the 1.88 Ga Post-Waterberg pole from the Kaapvaal craton (Hanson et al., 2004), Hanson et al. (2011) proposed that Kalahari did not assemble until after 1.87 Ga. However the quality of the Post-Waterberg pole is questionable as it is not supported by a field test nor a reversal test. We therefore keep Kalahari as a unified craton in the 1.87 Ga reconstruction.

6. Conclusions

In this paper we present a new U-Pb (zircon) age of 1870 ± 9 Ma for a reversed-polarity dyke and a mean Pb-Pb (zircon) age of 1867 ± 8 Ma for a normal-polarity dyke from Keuruu, Central Finland. The paleomagnetic results reveal a dual-polarity magnetization with a pronounced reversal asymmetry, which we explain mainly by an unremoved secondary component contaminating both N and R polarity components. Rock magnetic, peterophysical and geochemical data are similar between N and R dykes. By combining the dual polarity means, we obtained a key pole for Fennoscandia at Plat = 45.7°N, Plon = 230.9°E, K = 34.7, A95 = 5.5°. The ChRM is carried by magnetite, most of which is in the form of primary titanomagnetite grains crystallized from basaltic magma. The primary nature of the ChRM is further supported by positive baked contact tests. As the normal polarity dykes and unbaked host rock show similar paleomagnetic directions, we illustrated the baked contact test by means of the behavior of the decay of remanent magnetization intensity, which supports that the N polarity dykes also have a primary ChRM. Our data positions Baltica at low latitudes (19°), and aligns with Superior in a way that would smoothly proceed into the 1.8–1.2 Ga NENA (North Europe–North America) configuration. Paleosecular variation analysis of N polarity dykes is consistent with the low latitude position of Baltica.

Acknowledgements

Support for this work was provided by the The Doctoral Program of Geosciences of the University of Helsinki, the Vilho, Yrjö and Kalle Väisälä Foundation. We would like to thank Satu Mertanen, Hannu Huhma, Mikko Nironen, Yann Lahaye, and Hugh O'Brien, from the Geological Survey of Finland, and Ilmo Kukkonen, and Johanna Salminen from the University of Helsinki, for their support in this work. A special thanks to the reviewers for their constructive comments, which helped to improve this paper significantly.

Appendix A

See Tables A1 and A2.

Table A1
U–Pb age data of sample A2407 Saukko (dyke R2).

Sample information						Isotopic ratios						r	Disc.	Apparent ages (Ma)					
Sample/spot #	Analyzed zircon domainin BSE/CL image	U (ppm)	^{206}Pb (ppm)	^{206}Pb/^{204}Pb (measured)	f_{206}%	^{207}Pb/^{206}Pb	±s (%)	^{207}Pb/^{235}U	±s (%)	^{206}Pb/^{238}U	±s (%)		%	^{207}Pb/^{206}Pb	±s	^{207}p/^{235}U	±s	^{206}p/^{238}U	±s
A2407-01a	Euhedral, zoned, outer domain	476	162	5.91E+04	0.00	0.1148	0.0006	5.504	0.295	0.3478	0.0185	1.00	3.0	1876	9	1901	46	1924	89
A2407-01b	Euhedral, zoned, inner domain	1027	346	1.75E+05	0.00	0.1142	0.0006	5.442	0.289	0.3457	0.0183	1.00	2.9	1867	10	1892	46	1914	88
A2407-02a	Euhedral, CL-pale homog outer domain	1659	545	2.82E+05	0.00	0.1140	0.0006	5.343	0.282	0.3400	0.0178	0.99	1.4	1864	10	1876	45	1886	86
A2407-02b	Euhedral, CL-darker homog inner domain	414	136	3.44E+04	0.10	0.1144	0.0007	5.368	0.282	0.3402	0.0177	0.99	1.0	1871	11	1880	45	1888	85
A2407-03a	Euhedral, CL-pale homog	443	149	4.25E+04	0.04	0.1143	0.0006	5.513	0.290	0.3497	0.0183	0.99	4.0	1869	10	1903	45	1933	87
A2408-01a	Euhedral, CL-pale hazy	583	181	4.36E+04	0.10	0.1120	0.0006	4.684	0.282	0.3034	0.0182	1.00	-7.7	1832	10	1764	50	1708	90

Abbreviations used: homog. = homogeneous texture; osc. = oscillatory zoned; metam. = supposedly metamorphic/recrystallized/homogenized zircon domain; f_{206}, percentage of common ^{206}Pb in measured ^{206}Pb, calculated from the ^{204}Pb signal assuming a present-day Stacey and Kramers (1975) model terrestrial Pb-isotope composition; r, error correlation in conventional concordia space; Disc., Age discordance at closest approach of 1 sigma error ellipse to concordia.

Table A2
U–Pb age data of sample A902 Pirttivuori (dyke PK).

Sample information						Isotopic ratios						r	Disc. %	Apparent ages (Ma)					
Sample/spot #	Analyzed zircon domainin BSE/CL image	U (ppm)	^{206}Pb (ppm)	^{206}Pb/^{204}Pb (measured)	f_{206}%	^{207}Pb/^{206}Pb	±s (%)	^{207}Pb/^{235}U	±s (%)	^{206}Pb/^{238}U	±s (%)			^{207}Pb/^{206}Pb	±s	^{207}P/^{235}U	±s	^{206}P/^{238}U	±s
A902-42a	Euhedral, CL-bright, osc. Zoned	590	124	3.60E+04	0.00	0.0906	0.0005	3.286	0.077	0.2632	0.0060	0.97	5.3	1438	10	1478	18	1506	31
A902-46a	Euhedral, CL-pale, osc. Zoned	431	130	5.62E+02	2.70	0.1077	0.0009	5.506	0.154	0.3709	0.0100	0.96	18.1	1760	14	1902	24	2034	47
A902-39a	Euhedral, CL-bright, fairly homog. (BSE-dark)	55	13	1.08E+04	0.00	0.1091	0.0007	4.809	0.210	0.3197	0.0138	0.99	0.3	1784	11	1786	37	1788	68
A902-39b	Euhedral, CL-bright, osc.zoned (BSE-dark)	108	25	1.13E+04	0.00	0.1102	0.0006	4.756	0.205	0.3131	0.0134	0.99	−3.0	1802	10	1777	36	1756	66
A902-46b	Euhedral, CL-pale, osc. Zoned	466	143	9.64E+02	1.70	0.1107	0.0008	5.906	0.164	0.3871	0.0104	0.96	19.4	1811	13	1962	24	2109	48
A902-43a	Euhedral, CL-dark homog	793	232	1.21E+05	0.02	0.1141	0.0007	5.763	0.202	0.3665	0.0127	0.98	9.3	1865	11	1941	30	2013	60
A902-41a	Rounded, CL-dark homog./metam.	1266	353	1.63E+05	0.01	0.1142	0.0006	5.492	0.134	0.3489	0.0083	0.98	3.9	1867	10	1899	21	1929	40
A902-07a	Subhedral, CL-dark, homog.	827	254	8.19E+04	0.00	0.1142	0.0006	5.593	0.274	0.3551	0.0173	0.99	5.6	1868	9	1915	42	1959	82
A902-10a	Euhedral, osc. zoned	956	284	9.33E+04	0.00	0.1142	0.0006	5.458	0.256	0.3465	0.0161	0.99	3.1	1868	10	1894	40	1918	77
A902-17a	Euhedral, CL-dark homog./metam.	1137	321	1.38E+05	0.00	0.1143	0.0006	5.597	0.244	0.3553	0.0154	0.99	5.7	1868	9	1916	38	1960	73
A902-63a	Rounded, CL-dark homog./metam.	341	96	2.87E+04	0.09	0.1142	0.0008	5.568	0.328	0.3535	0.0207	0.99	5.2	1868	12	1911	51	1951	98
A902-23a	Euhedral, CL-dark homog.	1537	427	9.27E+04	0.00	0.1145	0.0006	5.754	0.270	0.3647	0.0170	0.99	8.3	1871	10	1940	41	2004	80
A902-25a	Rounded, CL-bright/ metam.	1390	351	2.13E+05	0.00	0.1144	0.0007	5.318	0.239	0.3371	0.0151	0.99	0.1	1871	10	1872	38	1873	73
A902-47a	Euhedral, BSE/CL-dark homog.	9	2	7.01E+02	0.99	0.1145	0.0020	5.194	0.567	0.3289	0.0354	0.99	−2.4	1872	31	1852	93	1833	172
A902-53a	Rounded, CL-dark homog./metam.	1449	400	1.69E+05	0.00	0.1146	0.0008	5.628	0.174	0.3562	0.0108	0.98	5.6	1873	11	1920	27	1964	51
A902-06a	Euhedral, CL-dark osc. zoned	462	145	1.55E+04	0.09	0.1146	0.0006	5.739	0.287	0.3632	0.0180	0.99	7.7	1874	9	1937	43	1997	85
A902-34a	Euhedral, CL-dark, homog.	692	195	4.78E+04	0.01	0.1147	0.0006	6.007	0.304	0.3799	0.0191	0.99	12.5	1875	10	1977	44	2076	89
A902-04a	Euhedral, CL-dark osc. zoned	328	105	4.36E+04	0.00	0.1148	0.0006	5.705	0.279	0.3605	0.0176	0.99	6.7	1877	9	1932	42	1984	83
A902-61a	Euhedral, CL-dark weakly zoned-homog.	286	83	4.15E+04	0.04	0.1149	0.0008	5.708	0.344	0.3603	0.0215	0.99	6.5	1878	12	1933	52	1984	102
A902-08a	Euhedral, osc. zoned	614	192	1.88E+04	0.25	0.1149	0.0007	5.717	0.281	0.3607	0.0176	0.99	6.6	1879	10	1934	43	1986	84
A902-49a	Euhedra, CL-dark homog.inner domain/ metam.	2142	589	6.05E+05	0.01	0.1149	0.0008	5.590	0.193	0.3527	0.0120	0.98	4.2	1879	12	1915	30	1948	57
A902-50a	Euhedra, CL-dark homog.inner domain/ metam.	704	196	6.88E+04	0.00	0.1150	0.0007	5.687	0.141	0.3588	0.0087	0.97	6.0	1879	10	1929	21	1977	41
A902-19a	Euhedral, osc. zoned	190	49	2.09E+04	0.00	0.1150	0.0007	5.200	0.223	0.3280	0.0139	0.99	−3.1	1880	10	1853	36	1829	68
A902-33a	Euhedral, CL-dark, homog.	624	172	1.04E+05	0.00	0.1150	0.0006	5.920	0.271	0.3733	0.0170	0.99	10.2	1880	9	1964	40	2045	80
A902-44a	Euhedral, CL-dark homog	469	127	5.37E+04	0.00	0.1150	0.0006	5.401	0.134	0.3406	0.0082	0.98	0.6	1880	10	1885	21	1889	40
A902-09a	Euhedral, osc. zoned	184	59	3.96E+04	0.01	0.1151	0.0006	5.808	0.273	0.3659	0.0171	0.99	7.9	1882	9	1948	41	2010	81
A902-20a	Euhedral, CL-dark, weakly zoned	403	115	4.91E+04	0.00	0.1151	0.0006	5.787	0.257	0.3646	0.0160	0.99	7.5	1882	10	1945	38	2004	76

(*continued on next page*)

Table A2 (*continued*)

Sample information						Isotopic ratios						r	Disc. %	Apparent ages (Ma)					
Sample/spot #	Analyzed zircon domainin BSE/CL image	U (ppm)	^{206}Pb (ppm)	^{206}Pb/^{204}Pb (measured)	f_{206}%	^{207}Pb/^{206}Pb	±s (%)	^{207}Pb/^{235}U	±s (%)	^{206}Pb/^{238}U	±s (%)			^{207}Pb/^{206}Pb	±s	^{207}P/^{235}U	±s	^{206}P/^{238}U	±s
A902-45a	Euhedral, weakly osc. Zoned	456	125	7.20E+04	0.08	0.1152	0.0006	5.541	0.144	0.3490	0.0089	0.98	2.9	1882	10	1907	22	1930	42
A902-11a	Euhedral, CL-dark homog.	450	131	1.87E+04	0.08	0.1152	0.0007	5.660	0.260	0.3563	0.0162	0.99	5.0	1883	10	1925	40	1965	77
A902-22a	Subhedral, CL-homog./ metam.	474	125	3.95E+04	0.00	0.1152	0.0006	5.465	0.254	0.3441	0.0159	0.99	1.4	1883	10	1895	40	1906	76
A902-12b	Euhedral, CL-dark osc. zoned center	348	101	3.43E+04	0.01	0.1154	0.0007	5.874	0.280	0.3693	0.0175	0.99	8.7	1885	10	1957	41	2026	82
A902-51a	Euhedral, CL-bright	79	21	1.68E+04	0.00	0.1153	0.0007	5.561	0.159	0.3497	0.0098	0.98	3.0	1885	12	1910	25	1933	47
A902-29a	Euhedral, osc. zoned	375	101	2.54E+04	0.00	0.1155	0.0007	5.850	0.271	0.3674	0.0169	0.99	8.0	1888	10	1954	40	2017	80
A902-32a	Euhedral, CL-pale-bright, zoned (stripes)	122	31	1.53E+04	0.00	0.1155	0.0006	5.358	0.228	0.3364	0.0142	0.99	−1.1	1888	10	1878	36	1869	68
A902-16a	Euhedral, CL-pale (BSE-dark)	176	47	2.26E+04	0.00	0.1156	0.0007	5.349	0.231	0.3357	0.0144	0.99	−1.4	1889	10	1877	37	1866	69
A902-30a	Euhedral, osc. zoned	369	96	3.35E+04	0.00	0.1156	0.0007	5.672	0.242	0.3558	0.0150	0.99	4.5	1889	10	1927	37	1962	71
A902-37a	Euhedral, osc. zoned	453	113	5.07E+04	0.00	0.1156	0.0006	5.406	0.235	0.3391	0.0146	0.99	−0.4	1889	9	1886	37	1882	70
A902-38a	Euhedral, CL-dark homog.	289	75	3.90E+04	0.00	0.1156	0.0006	5.787	0.261	0.3632	0.0163	0.99	6.7	1889	10	1944	39	1997	77
A902-52a	Rounded, CL-bright spotty	103	28	1.28E+04	0.00	0.1156	0.0007	5.613	0.142	0.3523	0.0087	0.97	3.5	1889	11	1918	22	1945	41
A902-36a	Euhedral, osc. zoned	457	112	3.31E+04	0.00	0.1158	0.0008	5.359	0.273	0.3357	0.0169	0.99	−1.6	1892	11	1878	44	1866	82
A902-48a	Rounded, CL-pale, zoned/metam.	119	33	1.29E+04	0.00	0.1158	0.0007	5.743	0.144	0.3598	0.0088	0.97	5.5	1892	10	1938	22	1981	42
A902-27a	Euhedral, osc. zoned	205	51	3.21E+04	0.00	0.1159	0.0007	5.376	0.242	0.3365	0.0150	0.99	−1.4	1893	10	1881	38	1870	72
A902-28a	Subhedral, CL-spotty, / metam.	201	50	2.04E+04	0.00	0.1159	0.0007	5.401	0.242	0.3381	0.0150	0.99	−1.0	1893	10	1885	38	1878	72
A902-15a	Euhedral, osc. zoned	291	85	2.82E+04	0.00	0.1162	0.0007	5.957	0.279	0.3718	0.0173	0.99	8.5	1899	10	1970	41	2038	81
A902-35a	Euhedral, osc. zoned	166	44	1.68E+04	0.00	0.1162	0.0007	5.733	0.274	0.3579	0.0170	0.99	4.5	1899	10	1936	41	1972	81
A902-31a	Rounded, metam.	225	65	2.13E+04	0.01	0.1164	0.0006	6.062	0.271	0.3777	0.0167	0.99	10.1	1902	9	1985	39	2065	78
A902-12a	Euhedral, CL-dark homog. Tip	399	110	3.40E+04	0.00	0.1167	0.0007	5.512	0.251	0.3426	0.0155	0.99	−0.4	1906	10	1903	39	1899	74
A902-62a	Euhedral, CL-dark weakly zoned	255	82	3.51E+04	0.00	0.1185	0.0017	6.552	0.413	0.4011	0.0246	0.98	14.7	1933	23	2053	56	2174	113
A902-54a	Euhedra, CL-dark homog.inner domain/ metam.	476	145	4.81E+04	0.00	0.1274	0.0008	6.886	0.225	0.3920	0.0125	0.98	4.0	2063	11	2097	29	2132	58
A902-13a	Euhedral, CL-dark homog.	310	129	3.53E+04	0.00	0.1773	0.0012	12.524	0.656	0.5122	0.0266	0.99	1.8	2628	11	2645	49	2666	113
A902-21a	Rounded, CL-dark, homog.	361	141	3.57E+04	0.00	0.1776	0.0011	11.952	0.631	0.4882	0.0256	0.99	−3.1	2630	10	2601	49	2563	111
A902-14a	Euhedral, CL-dark homog./metam.	628	253	9.30E+04	0.00	0.1780	0.0011	12.140	0.606	0.4948	0.0245	0.99	−2.0	2634	10	2615	47	2591	106
A902-05a	Euhedral, CL-dark osc. zoned	435	197	6.37E+04	0.00	0.1781	0.0011	12.385	0.657	0.5042	0.0266	0.99	−0.2	2636	10	2634	50	2632	114
A902-60a	Rounded, CL-pale homog./metam.	94	40	1.28E+04	0.21	0.1781	0.0014	12.534	0.818	0.5103	0.0331	0.99	1.0	2636	13	2645	61	2658	141
A902-03a	Euhedral, CL-dark homog.	437	198	5.19E+04	0.00	0.1786	0.0011	12.280	0.657	0.4988	0.0265	0.99	−1.4	2639	10	2626	50	2609	114
A902-57a	Rounded, CL-pale homog./metam.	90	39	3.61E+04	0.00	0.1786	0.0014	12.569	0.895	0.5103	0.0361	0.99	0.8	2640	13	2648	67	2658	154
A902-18a	Rounded, CL-dark homog./metam.	424	176	2.43E+05	0.00	0.1788	0.0011	12.637	0.610	0.5125	0.0245	0.99	1.2	2642	10	2653	45	2667	105
A902-24a	Subhedral, CL-dark, homog.	446	173	5.99E+04	0.00	0.1791	0.0011	12.428	0.633	0.5033	0.0254	0.99	−0.8	2644	10	2637	48	2628	109

Table A2 (*continued*)

Sample information						Isotopic ratios					r	Disc.	Apparent ages (Ma)					
Sample/spot #	Analyzed zircon domainin BSE/CL image	U (ppm)	^{206}Pb (ppm)	^{206}Pb/^{204}Pb (measured)	f_{206}%	^{207}Pb/^{206}Pb	±s (%)	^{207}Pb/^{235}U	±s (%)	^{206}Pb/^{238}U	±s (%)		^{207}Pb/^{206}Pb	±s	^{207}p/^{235}U	±s	^{206}p/^{238}U	±s
A902-56a	Rounded, CL-bright homog./metam.	46	21	3.01E+04	0.00	0.1791	0.0014	12.915	0.957	0.5229	0.0385	0.99	3.1	2645	13	2673	70	2711	163
A902-01a	Rounded, sector zoned Cl-pale/metam.	135	59	2.09E+04	0.00	0.1798	0.0011	11.822	0.622	0.4768	0.0249	0.99	−6.3	2651	10	2590	49	2513	109
A902-02a	Rounded, CL-bright/metam.	82	35	1.18E+04	0.00	0.1819	0.0012	11.591	0.603	0.4622	0.0239	0.99	−9.9	2670	10	2572	49	2449	105
A902-59a	Rounded, CL-bright homog.t/metam.	34	15	7.57E+03	0.00	0.1823	0.0015	13.568	0.947	0.5396	0.0374	0.99	5.0	2674	13	2720	66	2782	157
A902-55a	Rounded, CL-bright homog./metam.	36	15	7.04E+03	0.00	0.1831	0.0014	12.638	0.926	0.5005	0.0365	0.99	−3.0	2682	12	2653	69	2616	157
A902-40a	Euhedral, CL-bright (BSE-dark)	64	24	5.05E+03	0.00	0.1835	0.0011	12.814	0.602	0.5065	0.0236	0.99	−1.9	2685	10	2666	44	2642	101
A902-26a	Euhedral, CL-dark homog.	40	16	5.35E+03	0.00	0.1850	0.0013	12.970	0.654	0.5085	0.0254	0.99	−2.2	2698	11	2677	48	2650	108
A902-58a	Rounded, CL-bright zoned/metam.	80	35	1.62E+04	0.00	0.1885	0.0017	13.542	0.887	0.5210	0.0338	0.99	−1.2	2729	14	2718	62	2703	143

Abbreviations used: homog. = homogeneous texture; osc. = oscillatory zoned; metam. = supposedly metamorphic/recrystallized/homogenized zircon domain; f_{206}, Percentage of common ^{206}Pb in measured ^{206}Pb, calculated from the ^{204}Pb signal assuming a present-day Stacey and Kramers (1975) model terrestrial Pb-isotope composition; r, error correlation in conventional concordia space; Disc., Age discordance at closest approach of 1 sigma error ellipse to concordia.

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